Artifact: A Reproducible, Database-Free Pipeline for Detecting Research-Software Paradigms in the DELFI Community

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Abstract: This artifact is a *dataset publication* accompanying the DELFI 2026 paper *Research Software Engineering in the Learning Technologies: a replication study of research paradigms in the DELFI community*. It releases the full set of intermediate datasets behind the study — the LLM annotation runs, the within- and cross-model aggregated labels, the human re-annotations, the reference-study evaluation, and the derived figures and tables — together with the pipeline that produces them. Every published number and figure is reproducible from the shipped datasets alone, *without* the copyrighted DELFI paper PDFs or a commercial LLM API; the steps that consume those sources are optional, because their outputs are themselves published as datasets. A programmatic guard enforces that no paper full text is ever contained in the released data. We describe the released datasets, the pipeline that regenerates them, the reproducibility guarantees, and the constraints that motivated the design for the artifact-evaluation track.

Resource Type: Data set (with an accompanying software pipeline)
License: Code: MIT; Data: CC-BY-4.0
Stable ID: *archival DOI to be minted on Zenodo before camera-ready* (e.g. `10.5281/zenodo.XXXXXXX`)
Repository: https://github.com/juliandehne/rse_in_delfi_artifact

Keywords: Research Software Engineering, replication, large language models, inter-coder reliability, reproducibility, DELFI

1 Introduction

The companion study [DLW26a] replicates and extends a 2017 analysis of research-software paradigms across the DELFI proceedings, using three open-weight large language models (LLMs) to annotate $n = 798$ papers and validating the result against the earlier human-coded reference study. This artifact packages the *evidence and the computation* behind those results so that a reviewer can re-run the analysis end-to-end on a laptop in minutes.

The central design constraint is legal: the analyzed DELFI papers are copyrighted and may not be redistributed. The artifact therefore ships only derived data (LLM labels, justifications, human re-annotations and bibliographic metadata) and is explicitly built so that the two paper-dependent steps and the API-dependent step can be skipped while still reproducing every published number and figure.

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2 Artifact overview

The primary artifact is the dataset collection under `data/` (enumerated in the abstract). Accompanying it are a Python package (`pipeline/`: `config`, the six steps and a full-text guard) and the management script `run_pipeline.py`, which regenerate those datasets and figures from one another; every published number and figure is reproducible from the shipped datasets alone. We target the data *Available* and *Reusable* badges: the datasets are archived, openly licensed, documented with a data dictionary, and accompanied by the code that regenerates them.

3 The pipeline

The pipeline has six steps (Table 1); each step's output is a published dataset. Steps 3, 5 and 6 regenerate their datasets from the shipped data and run by default. Steps 1, 2 and 4 are *optional*: their outputs are already published, so they need only be re-run to rebuild those datasets from the original paper PDFs (steps 1 and 4) or the LLM API (step 2).

Tab. 1: Pipeline steps and default execution policy.

#	Step	Default	Requires
1	Preprocessing (PDF → text)	**optional**	paper PDF folder
2	LLM experiments	**optional**	SAIA/KISSKI API token
3	Descriptive analysis (figures)	runs	shipped data
4	Human re-annotation	**optional**	paper PDF folder (interactive)
5	Label aggregation (intra + inter, ICR)	runs	shipped data
6	Secondary analysis (paradigm biases)	runs	shipped data

Data flow. Step 1 extracts text from the PDFs; step 2 annotates each paper with three LLMs (three runs each) via the SAIA API. Step 5a aggregates the runs per model (majority vote) with intra-model ICR; step 4 elicits a human label where all three models disagree; step 5b aggregates across models (majority vote, ties broken by the human label), reports cross-model ICR and validates against the 2017 reference study. Steps 3 and 6 produce the figures and the bias analysis. `run_pipeline.py` runs the default steps; `--paper-folder` enables steps 1/4 and `--run-experiments` (with a SAIA token) enables step 2. The no-full-text guard runs before and after every invocation.

4 Reproducing the results

After `pip install -r requirements.txt`, `python run_pipeline.py` regenerates — from the shipped data, with no PDFs or API key — the final aggregated dataset (798 × 185), the ICR tables, the paradigm-distribution and label-trend figures, and the combined-label bias tables. The recomputed cross-model inter-coder reliability is Krippendorff's α [Kr04] (binary labels complemented by Gwet's AC1 [Gw08]): average $\alpha = 0.51$ (binary), 0.66 (ordinal) and 0.60 overall, with $\alpha = 0.33$ against the 2017 reference study. The three models disagreed completely on 0, 6, 17 and 9 papers for the process, outcome, design and acceptance paradigms; those cases were resolved with the shipped human labels.

5 Data and the no-full-text guarantee

The released data contain only labels, LLM/human justifications and bibliographic metadata. The original annotation script dropped the `abstract`, `text` and `references` columns before persisting results "for legal reasons"; the artifact preserves and enforces this invariant programmatically: `pipeline/fulltext_guard.py` scans every shipped table and fails if a banned full-text column is present or any cell exceeds a length threshold incompatible with a short justification. Step 1's full-text output goes only to a `.gitignore`'d directory.

6 LLM documentation

Per the DELFI AET LLM policy, Table 2 lists the three open-weight models used in step 2, all served through the KISSKI SAIA API. Each model was run three times (run_1–3). All runs used identical, deterministic inference parameters: **temperature** = 0, **top_p** = 1.0, **seed** = 42, no **stop sequences**, no explicit **max-tokens** cap (provider default), a 300 s request timeout, and no function/tool calling (plain JSON-mode prompting). The models are open-weight, so a closed-weight snapshot date does not apply; the snapshot is pinned by the version in the model identifier, and the run dates are recorded in the result filenames.

Tab. 2: LLM models and versions used for annotation (KISSKI SAIA API).

Short name	Model identifier (version)	Runs
mistral	`mistral-large-3-675b-instruct-2512`	3
llama	`llama-3.1-sauerkrautlm-70b-instruct`	3
gemma	`gemma-3-27b-it`	3

Prompts. The complete prompt — the system prompt and the user-prompt template with its `{row[...]}` placeholders — is shipped verbatim at `data/prompt_templates/prompt_template_1.md`. The placeholders `title`, `authors` and `year` resolve to public bibliographic metadata; the placeholders `abstract`, `text` and `references` resolve to the copyrighted paper content and are therefore the *only* part of a fully resolved prompt that cannot be redistributed. The substitution logic (`pipeline/step2_experiments.py`, `fill_user_prompt`) is included so the resolved prompt can be regenerated by anyone holding the paper PDFs.

7 FAIR preparation and targeted badges

The artifact targets the **Available** and **Reusable** badges: *Findable/Available* via a stable archival DOI (Zenodo, from a tagged release) and machine-readable metadata (`CITATION.cff`); *Accessible* via open licensing (MIT code, CC-BY-4.0 data; `LICENSE`, `LICENSE-data`); *Interoperable* via open CSV with a documented data dictionary (`data/README.md`); and *Reusable* via a parameterised, documented, re-runnable pipeline. No personal or human-subject data are released, so no further DSGVO/ethics statement beyond the no-full-text guarantee is required.

8 Requirements and environment

Python 3.11+, `pandas`, `numpy`, `scikit-learn`, `krippendorff`, `statsmodels`, `matplotlib` and `openpyxl`; see `requirements.txt`. `pymupdf` is needed only to re-run step 1, and an OpenAI-compatible client plus a SAIA token only to re-run step 2.

9 Limitations

Steps 1 and 2 cannot be exercised by reviewers without the copyrighted PDFs and an API token; their outputs are shipped and verified downstream instead. LLM annotations are sampled at temperature 0 with a fixed seed, but exact reproducibility of step 2 ultimately depends on the provider's model snapshots.

References

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